HoQ & FMECA Analysis of a Robotic Vacuum Cleaner

Project Report

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1 Introduction

The increasing integration of smart technologies into domestic environments has significantly reshaped consumer expectations regarding home maintenance and household automation. Among these innovations, robot vacuum cleaners have emerged as a leading category within the smart home ecosystem, driven by advances in artificial intelligence, sensor technology, and user-friendly digital interfaces.

This report focuses on the conceptualization and early design stages of a smart robot vacuum cleaner specifically engineered for families with children and pets; a demographic whose lifestyle characteristics create unique cleaning and hygiene challenges.

Families with young children and companion animals typically engage in frequent cleaning due to the accumulation of food particles, accidental spills, tracked-in dirt, pet hair, and airborne allergens. From a user-experience perspective, parents in such households often navigate densely scheduled routines involving school commutes, professional responsibilities, and childcare demands. Consequently, they exhibit a high demand for automated cleaning solutions that reduce manual cleaning time and facilitate a healthier living environment. This project positions the robot vacuum as a technologically advanced yet accessible tool to support these families by offering enhanced suction performance, intelligent navigation, real-time app connectivity, and operational features optimized for safe use around children and pets.

2 Market Segment Determination

The process of determining the primary market segment involved examining demographic characteristics, behavioural patterns, and contextual factors influencing purchasing decisions (Maisano, 2025c). For this study, the market segment is defined as modern families with children (ages 0–12) and one or more household pets, such as primarily dogs and cats. This segment exhibits distinctive functional requirements and value expectations compared to other potential user groups, such as single professionals, elderly individuals, or technology enthusiasts.

2.1 Demographic and Household Characteristics

Families within this segment typically:

- Reside in medium to large homes with multiple surface types (e.g., hardwood floors, carpets, tiles).
- Have at least one child whose activities generate frequent small-scale messes.
- Own one or more pets that shed hair and introduce external dirt into the home.

These households prioritize hygiene and allergen reduction due to the increased sensitivity of children to environmental contaminants, making them highly receptive to reliable, automated cleaning solutions.

2.2 Behavioural and Psychographic Traits

This market segment tends to value:

- Convenience and time savings, as parents balance complex schedules.
- Technological integration, including Wi-Fi-enabled devices and smart home ecosystems.
- Reliability and low maintenance, to avoid disruptions in their daily routine.
- Safety and quiet operation, necessary in environments with babies, toddlers, and pets that may react adversely to loud or unpredictable machines.

These psychographic traits suggest a strong alignment with smart home appliances offering autonomy, customization, and high usability.

2.3 Drivers of Product Adoption

Adoption of a robot vacuum cleaner in this target segment is influenced by several key drivers:

- **Increased cleaning frequency:** Daily dirt accumulation from both children and pets leads families to seek devices capable of performing multiple cleaning cycles per day without supervision.
- **Health and hygiene concerns:** Allergens, pet dander, and food residues are critical concerns for parents, making suction power, filtration quality, and thorough area coverage essential.
- **Preference for automation:** Families adopt technologies that reduce cognitive load. The ability to schedule cleaning sessions or control the device remotely aligns with this preference.
- **Growing familiarity with smart home technologies:** Households are increasingly accustomed to smart speakers, cameras, thermostats, and lighting systems. This familiarity lowers adoption barriers for smart cleaning devices.

2.4 Market Potential

The segment's potential is reinforced by global trends, including expanding pet ownership rates, greater emphasis on household hygiene post-pandemic, and rising adoption of connected home solutions (Intelligence, 2024; Statista, 2025) This makes families with children and pets an attractive and strategically aligned target segment for a robot vacuum cleaner with enhanced performance and smart capabilities.

3 House of Quality (HoQ)

3.1 Customer Requirements (CR) Collection

A set of Voice of the Customer (VoC) statements was obtained. Some representative comments from the target audience included:

- "It has to clean well on every surface, including carpets and hard floors."
- "It should be easy to use for everyone in the family."
- "It should be quiet and not disturb children or pets."

- "I would like to control it and see information from an app."

The answers were collected and analysed to identify recurring themes and expectations regarding a large robotic vacuum cleaner for families with children and pets.

Starting from this initial set of VoCs, an iterative refinement process was carried out to remove duplicates, merge similar statements, and retain only the most representative needs. This process resulted in a preliminary list of customer requirements, which was further refined using the Customer Requirement Tree.

3.2 CR Tree & Clustering

The Customer Requirement Tree was used to translate the VoCs into actionable customer requirements. This tool provides a clearer picture of customer needs by structuring them from general expectations into more specific and measurable requirements relevant to the final product.

Figures 4-6 in Appendix A, illustrate the path from the initial VoCs to the final CRs. After freezing the final requirements, they were clustered into categories.

Table 1 presents the final list of 14 customer requirements for the robotic vacuum cleaner and their corresponding clustering.

Table 1: Final CRs with Clustering.

CR ID	CR	Category
1	Strong Suction Power	System Design Needs
2	Cleans All Floor Types	System Design Needs
3	Smart Navigation System	System Design Needs
4	Automatic Recharge & Resume	System Design Needs
5	Wide Area Coverage	System Design Needs
6	Energy Efficient and Eco-Friendly	System Design Needs
7	Low Noise Level	Comfort
8	Easy To Use (for All Users)	Comfort
9	Lightweight and Compact	Comfort
10	Long Component Life	Structural Needs
11	Mobile App Control	Additional Functionalities
12	Smart Mapping Visualisation	Additional Functionalities
13	Notifications and Alerts	Additional Functionalities
14	Efficient Cleaning Kit	Additional Functionalities

3.3 Prioritisation of CRs

Once the CRs had been defined and clustered, a survey, see Figures 7-9 in Appendix B was conducted to determine the priority assigned to each requirement by the market segment.

Correctly rating the importance of customer requirements is a crucial step, as it directly influences the target value setting of design requirements. Final importance ratings are typically determined by combining relative importance scores with competitive priority ratings. However, this task can be challenging due to uncertainty or imprecision in customer judgment.

In this HoQ, the median value of the survey responses was used as the key metric for the importance of each CR. Table 4 in Appendix C shows the CRs with their respective weights.

3.4 Products Benchmarking

To define the product's key differentiators and establish its competitive positioning, three competitors were identified. A competitive analysis of the CRs was conducted to identify areas for improvement and support the strategic development of the product roadmap. The selected products were the Dyson 360 Vis Nav (Amazon, 2024; Dyson, 2023; Elgiganten, 2025a), iRobot Roomba j7+ (iRobot, 2024; Elgiganten, 2025b), and Roborock S8 (Roborock, 2024; Elgiganten, 2025c). These competitors were chosen for their relevance to the same market segment: large robotic vacuums for families with children and pets. Furthermore, they exhibit characteristics and functionalities similar to the intended product and therefore provide useful insights for identifying improvement opportunities (Brian Bennett, 2021; David Ludlow, 2023; Jennifer Pattison Tuohy, 2024).

3.5 Competitive Benchmarking of CRs

The results of the competitive benchmarking of CRs are shown in Table 5 in Appendix D, but also summarised in the HoQ matrix, which shows the ratings for the proposed product and the three competitors listed above. These results were used to define the target satisfaction levels for each of the current CRs. For each requirement, an improvement ratio was calculated using Equation 1:

$$\text{Impr. Ratio} = \frac{\text{Target Level}}{\text{Current Level}} \tag{1}$$

After computing the improvement ratios, the next step involved identifying which areas, when improved, would most strongly contribute to competitive advantage and potential sales increases. To support this assessment, a relative weight - also referred to as a strength coefficient - was assigned to each CR.

These strength levels were translated into numerical coefficients and used, together with the CR importance values and the improvement ratios, to compute the absolute weight of each requirement. The following strength coefficients were adopted:

- Real: 1.5
- Potential: 1.2
- Uninfluential: 1.0

The absolute weight of each CR was calculated according to Equation 2:

$$\text{Abs. Weight} = \text{Importance} \times \text{Impr. Ratio} \times \text{Strength} \tag{2}$$

The absolute weights were then normalised to obtain the relative weights of the CRs, which are reported in the HoQ. These relative weights are used in subsequent sections to determine the most influential Technical Characteristics for the new robotic vacuum cleaner. Table 5 in Appendix **??** shows the result of the benchmarking.

3.6 Determination and Description of the Technical Characteristics (TCs)

The most influential TCs were selected to satisfy the CRs. This section provides a brief description of each TC and the formulas used to quantify them.

3.6.1 Motor Power

Motor power refers to the electrical power delivered by the device's main motor, measured in watts. It directly influences suction capability, mechanical performance, and overall operational efficiency. This parameter is essential for evaluating the system's ability to perform tasks involving mechanical force or suction. See Table 6 in Appendix E for the formula.

3.6.2 Airflow Rate

The airflow rate represents the volume of air that the device can move per second. This characteristic reflects the machine's ability to transport particles, remove debris, and support effective suction. A higher airflow rate improves cleaning efficiency and overall performance. See Table 6 in Appendix E for the formula.

3.6.3 SLAM Localization Accuracy

SLAM localization accuracy defines how precisely the system estimates its position using Simultaneous Localization and Mapping. Measured in centimetres, this parameter indicates the deviation between estimated and true positions. High accuracy enables reliable navigation, improved obstacle avoidance, and efficient path planning. See Table 6 in Appendix E for the formula.

3.6.4 SLAM Update Range

SLAM update range indicates the maximum distance at which the system can reliably update its map and localization data. Measured in metres, it reflects the effective detection range of sensors such as LiDAR or camera-based depth sensors. A longer update range improves navigation in larger or more complex environments. See Table 6 in Appendix E for the formula.

3.6.5 Path Optimization Metric

The path optimization metric measures the efficiency of the generated navigation route compared to an ideal or shortest path. A higher percentage indicates better optimization, leading to reduced travel time, lower energy consumption, and improved operational productivity. This metric is essential for evaluating autonomous decision-making and navigation algorithms. See Table 6 in Appendix E for the formula.

3.6.6 Auto Recharge Time

Auto recharge time refers to the duration needed to recharge the device's battery from a predefined low level to an operational level. Shorter recharge times reduce downtime and increase system availability. See Table 6 in Appendix E for the formula.

3.6.7 Battery Capacity

Battery capacity indicates the amount of electrical charge the battery can store. Higher capacity allows for longer operation between recharge cycles and increases system autonomy. See Table 6 in Appendix E for the formula.

3.6.8 Acoustic Noise Level

Acoustic noise level measures the emitted sound during operation, expressed in A-weighted decibels. Lower values increase user comfort and reduce environmental disturbance. See Table 6 in Appendix E for the formula.

3.6.9 Component Lifetime

Component lifetime refers to the expected operational duration of key mechanical and electronic components. A longer component lifetime reduces maintenance needs and enhances long-term value. See Table 6 in Appendix E for the formula.

3.6.10 Setting Steps

Setting steps represent the number of discrete adjustment levels available for system parameters. A higher number of steps enables finer control of performance characteristics. See Table 6 in Appendix E for the formula.

3.7 Prioritisation of TCs

After identifying the key TCs, ISM was applied to determine the absolute weight of each TC using Formula 3, as found in (Maisano, 2025c), presented below:

$$\text{Abs. Weight}(TC_j) = \sum_i [r_{ij} \cdot \text{Abs. Weight}(CR_i)] \tag{3}$$

The coefficient r_{ij} represents the relationship between $\mathrm{CR_i}$ and $\mathrm{TC_j}$, following the standard in (Maisano, 2025c). The values assigned are shown in the list below: (The signs indicating each value, in chronological order, can be seen in Figure 10 in Appendix F)

- 9 for a strong relationship
- 6 for a medium relationship
- 3 for a weak relationship

The relative weight of each TC was then calculated using Formula 4, as found in (Maisano, 2025c), presented below:

$$\text{Rel. Weight}(TC_j) = \frac{\text{Abs. Weight}(TC_j)}{\sum_j \text{Abs. Weight}(TC_j)} \tag{4}$$

3.8 Competitive Benchmarking of TCs

The benchmarking analysis aimed to evaluate the product's TCs against Dyson, iRobot, and Roborock to identify strengths, weaknesses, and areas for improvement.

The competitors were selected for their market alignment and features similar to those of our mid-range product segment. The selected competitors provided valuable insights into the expectations and benchmarks within the target market. The result of this benchmarking is presented in Table 2 below.

Table 2: Benchmarking Table for Technical Characteristics

Unit of Measurement	**W**	**m^3/s**	**cm**	**m**	**%**	**min**	**mAh**	**dB(A)**	**years**	**integer**
Absolute Weight (ISM)	2.27	3.85	1.98	2.68	3.34	3.46	4.85	2.27	4.68	1.10
Relative Weight (ISM)	7.44%	12.64%	6.48%	8.80%	10.96%	11.36%	15.92%	7.44%	15.36%	3.60%
Our Existing Model	2000	35	3	2.5	85	120	5200	70	3	4
Dyson's Model	2100	42	7.1	4	90	115	6000	76	4	3
iRobot's Model	1800	37	15	5	80	120	4200	70	4	3
Roborock's Model	2000	26	2	8	88	150	5200	68	4	3

The ISM weighting indicated that the two most influential TCs were Battery Capacity and Component Lifetime. These were followed by Airflow Rate, Auto Recharge Time, and Path Optimization Performance. These characteristics had the strongest influence on customer satisfaction and therefore played a central role in the evaluation.

Overall, the benchmarked model demonstrated solid performance in several medium- and high-importance areas. It exhibited competitive motor power, low acoustic noise levels, strong battery capacity, and reasonable recharge time, which placed it near the top of the benchmark in these categories. SLAM localization accuracy also performed well, although it remained below the level achieved by Roborock.

The analysis also identified several key weaknesses. The model showed significant underperformance in SLAM update range, where all competitors - particularly Roborock - demonstrated substantially superior capability. In addition, Component Lifetime, one of the most heavily weighted characteristics, was lower than all competitor values, making it a critical target for improvement. Airflow Rate was also below average, trailing both Dyson and iRobot, which reduced cleaning efficiency relative to the market leaders.

3.9 Lyman's Normalisation

When ISM was used to prioritise the TCs of the robot vacuum cleaner, the importance assigned to a group of TCs depended on the number of individual TCs included in that group. This introduced a bias whereby configurations containing a larger number of detailed features appeared more important, even when this did not accurately reflect customers' true preferences for robot vacuum cleaners.

To address this issue, Lyman proposed normalising the coefficients in the relationship matrix by dividing each coefficient by the sum of all coefficients in the corresponding row. The resulting adjusted coefficients were then used within the ISM procedure to determine the absolute weight of each TC (Maisano, 2025c).

The absolute weights of the TCs for the robot vacuum cleaner were computed accordingly and presented in Table 7 in Appendix G.

3.10 Q-Bench Algorithm

A Python script was developed to implement a decision-support procedure that combined the ELECTRE II outranking method with a Q-Bench-style local search, as recommended by (Maisano, 2025b). The objective was to identify the most advantageous configuration of TCs within predefined user limits.

3.10.1 Method Overview

The lower and upper bounds for each criterion were first defined, and the corresponding intervals were discretised into 21 equally spaced levels. This generated the search domain explored by the algorithm.

ELECTRE II was then applied to evaluate pairwise comparisons between solutions. For every pair, strong and weak concordance indices, the discordance index, and potential veto situations were computed. These indicators allowed the determination of whether one solution strongly outranked, weakly outranked, or did not outrank another.

3.10.2 Search Mechanism

Starting from the worst admissible solution (all criteria at their lower bounds), the algorithm iteratively tested small variations—one criterion at a time. A candidate solution was accepted only when ELECTRE II confirmed that it strongly or weakly outranked the current best solution. This process continued until no further improvement was possible, resulting in convergence toward a stable, locally optimal configuration.

3.10.3 Competitor Comparison

Once the optimal solution had been identified, it was compared with the competitors' profiles. If the optimised vector coincided with an existing alternative, the corresponding label was assigned; otherwise, the solution was considered unique.

3.10.4 Results

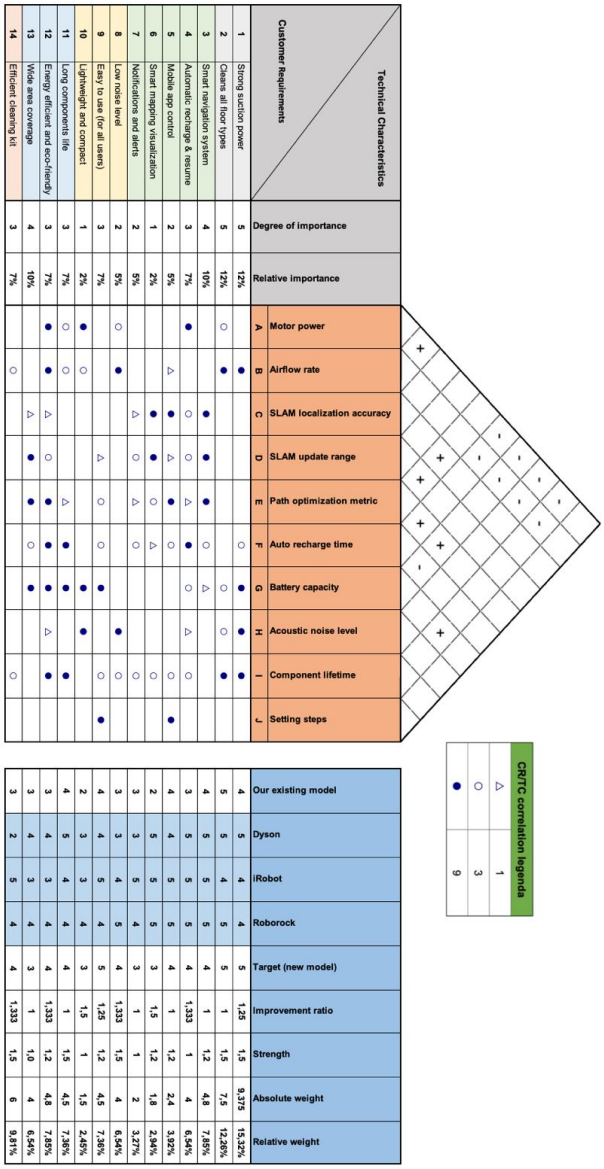
The optimisation procedure produced the following best solution within the allowed ranges:

$$[1600,\ 255,\ 30,\ 8,\ 100,\ 120,\ 8000,\ 60,\ 20,\ 1]$$

This configuration maximised performance according to the weighted criteria and ELECTRE II decision rules and did not correspond to any competitor's existing model. Figure 11 in Appendix H illustrates the profile of the optimised solution across all TCs.

4 House of Quality Final Design

Figure 1 below represents the final House of Quality assembly, showing all the different sections.



	Technical Characteristics / Customer Requirements	Degree of importance	Relative importance	A Motor power	B Airflow rate	C SLAM localization accuracy	D SLAM update range	E Path optimization metric	F Auto recharge time	G Battery capacity	H Acoustic noise level	I Component lifetime	J Setting steps	Our existing model	Dyson	iRobot	Roborock	Target (new model)	Improvement ratio	Strength	Absolute weight	Relative weight
1	Strong suction power	5	12%		●				○	●	●	●		4	5	4	4	5	1,25	1,5	9,375	15,32%
2	Cleans all floor types	5	12%	○	●					○	○	●		5	5	4	5	5	1	1,5	7,5	12,26%
3	Smart navigation system	4	10%			●	●	●		△				4	5	5	5	4	1	1,2	4,8	7,85%
4	Automatic recharge & resume	3	7%	●		○	○	△	●	○	△	○		3	5	5	5	4	1,333	1	4	6,54%
5	Mobile app control	2	5%		△	●	△	●	○			○	●	4	4	5	5	4	1	1,2	2,4	3,92%
6	Smart mapping visualization	1	2%			●	●	○	△			○		2	5	5	5	3	1,5	1,2	1,8	2,94%
7	Notifications and alerts	2	5%			△	○	△	○			○		3	3	5	4	3	1	1	2	3,27%
8	Low noise level	2	5%		●				○		●	○		3	3	4	5	4	1,333	1,5	4	6,54%
9	Easy to use (for all users)	3	7%	○			△	○	○	●		○	●	4	4	5	4	5	1,25	1,2	4,5	7,36%
10	Lightweight and compact	1	2%	●	○					●	●			2	3	3	4	3	1,5	1	1,5	2,45%
11	Long components life	3	7%	○	○					●		●		4	5	4	4	4	1	1,5	4,5	7,36%
12	Energy efficient and eco-friendly	3	7%	●	●	△	○		●	●	△	●		3	4	3	4	4	1,333	1,2	4,8	7,85%
13	Wide area coverage	4	10%			△	●		○	●				3	4	3	4	3	1	1,0	4	6,54%
14	Efficient cleaning kit	3	7%	○	○							○		3	2	5	4	4	1,333	1,5	6	9,81%
Total		41																				

CRTC correlation legenda	
△	1
○	3
●	9

Figure 1: House of Quality Final Design.

5 Failure Mode and Effects and Criticality Analysis Methodology

Failure Mode, Effects and Criticality Analysis (FMECA) is a systematic methodology used to identify and analyse potential failure modes within a system, together with their associated effects and causes. Its primary objective is to anticipate failures and reduce long-term risks by implementing preventive or corrective measures (Maisano, 2025a).

5.1 Failure Mode Identification

For each main component, the main functions and the parts responsible for delivering those functions were identified. Since a robot vacuum operates in several autonomous states, including navigation, obstacle avoidance, suction, and connectivity, the operational mode considered in this study encompassed all of these states.

Based on this functional breakdown, the failure modes corresponding to each function were identified. A failure mode is defined as the inability of a component or subsystem to meet its functional requirements. For the system under study, five representative failure modes were retained, see Table 3.

Table 3: FMECA Table.

Unique ref	Function	Operational mode	Failure mode(s)	Effect	S	Failure cause(s)	O	Detection system	D	RPN	Code
Sensor	Detect and avoid obstacles	Navigation / Obstacle avoidance	Obstacle sensor failure	Robot collides with objects or falls down stairs	5	Dust or dirt on lens; software miscalibration; firmware bug	7	Alarm from no signal	2	70	A1
Wheel/motor	Drive the robot's wheels	Movement / Locomotion	Wheel motor malfunction	Robot moves in circles or stops completely	6	Physical damage; hair wrapped around axle; wheel damage	8	Alarm from no signal; thermal resistor	9	432	B1
Battery	Provide sufficient power for operation	Energy supply / Charging	Battery degradation	Reduced autonomy; robot stops before cleaning is finished	6	Overheating; excessive charge/discharge cycles; high temperature exposure	7	Thermal resistor	2	84	C1
Bluetooth module	Send and receive data between robot and app	Connectivity / User interface	Communication loss	User cannot control or monitor the robot via the app	5	Faulty module; firmware incompatibility; signal interference	6	No Bluetooth signal	3	90	D1
Filter	Maintain airflow for debris collection	Cleaning	Filter clogged	Reduced suction power; debris not properly collected	3	Lack of maintenance; dusty environment; infrequent bin emptying	7	None	7	147	E3

For each failure mode, the associated effects were listed. These effects represent the potential consequences of the failure on the robot vacuum's performance, functionality, or safety.

The severity (S) column was then completed, reflecting the seriousness of the worst possible effect on the overall system. A severity rating of 1–2 corresponded to a minor effect, often unnoticed by the customer, whereas a severity rating of 10 represented a critical failure leading to major dissatisfaction or complete system shutdown.

5.2 Failure Cause Identification

To determine the root causes of each failure mode, typical failure mechanisms for robot vacuums were considered, including dust accumulation, overheating, electrical wear, sensor obstruction, and signal interference, as shown in Table 3.

For each failure cause, an occurrence (O) rating was assigned to represent the likelihood of the failure mechanism triggering the failure mode. A rating of 1–2 indicated an extremely low probability, whereas a rating of 9–10 indicated a very high probability, corresponding to frequent or recurring issues.

A detectability (D) rating was also assigned, based on the ability of the robot vacuum's internal systems to detect the failure before it affected the user. A rating of 1–2 meant the issue was highly likely to be detected early, while a rating of 10 indicated almost no chance of detection prior to user impact.

Detection systems considered in this study included internal sensors (such as pull-up transistors and thermal resistors), Bluetooth signal monitoring, and signal-loss detection.

5.3 FMECA Worksheet and Risk Priority Number (RPN)

The Risk Priority Number (RPN) quantified the criticality of each combination of failure mode and failure cause. It was calculated using Formula 5 below as presented in (Maisano,

2025a):

$$\text{RPN} = S \times O \times D \tag{5}$$

The RPN simultaneously accounted for the severity of the failure's consequence (S), the occurrence of the associated cause (O), and the detectability of the emerging issue (D). Higher RPN values indicated greater criticality, meaning that the corresponding failure mode–cause combination required priority attention in subsequent mitigation planning, as shown in Figure 2 below.

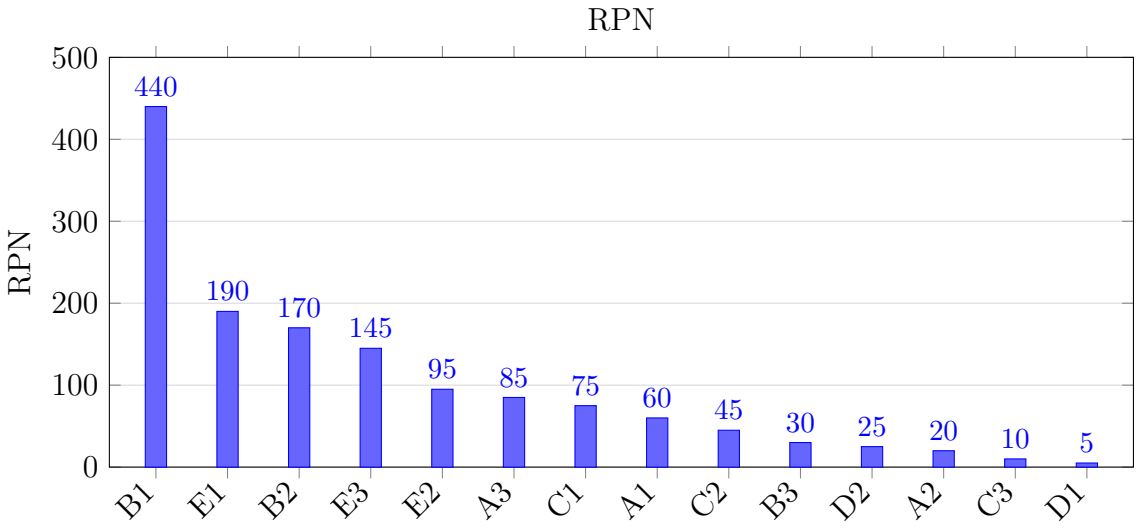


Figure 2: RPN Ranking.

5.4 Discussion of RPN Results

The analysis shows that failure modes such as wheel motor malfunction and filter clogging generated the highest RPN values due to their relatively high occurrence rates and substantial effects on system functionality. These combinations were therefore prioritised for further risk-mitigation measures, including improved protection, enhanced monitoring, or strengthened maintenance procedures.

The first priority concerned the combination labelled B1, corresponding to the failure cause "hair wrapped around the wheels" associated with the failure mode "wheel motor malfunction." This case required corrective actions due to its dominant RPN value. Subsequent priorities included the combinations E1, B2, E3, and etc.

5.5 Criticality Matrix

The risk associated with each failure mode resulted from the combination of its Severity and Occurrence ratings. These two parameters were visualized in a criticality matrix, with Severity plotted on one axis and Occurrence on the other. This matrix made it possible to quickly identify the failure modes presenting the highest operational risk.

Although the matrix focused on the S-O interaction, the Detectability rating also contributed to the overall criticality. For two failure modes with similar S and O values, the mode with a higher D rating was generally more problematic, as the failure was less likely to be detected before affecting the user. To convey this information visually, the size of each plotted point was made proportional to its Detectability value, allowing all three

criteria — Severity, Occurrence, and Detectability — to be interpreted simultaneously. The result is shown in Figure 3 below.

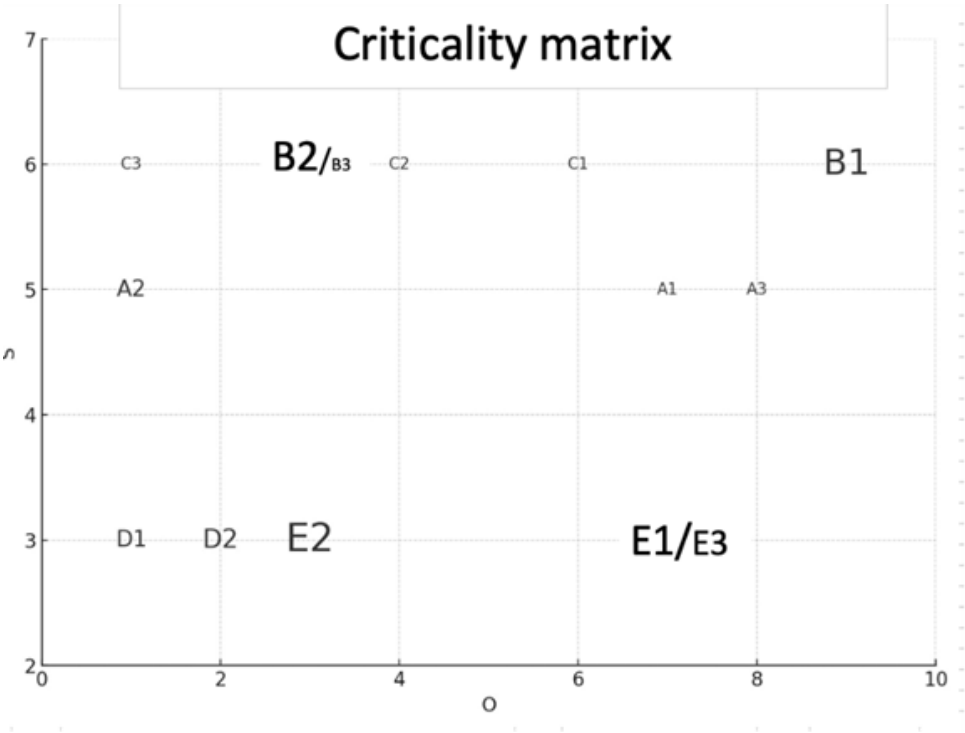


Figure 3: Criticality Matrix.

6 Discussion

This chapter reflects on the methodological choices, results, and limitations encountered throughout the project.

6.1 Market Segment

To gain a more accurate understanding of the target market, the group could have conducted interviews. Interviews often provide more valuable insights than surveys, as participants can articulate their needs, expectations, and frustrations in detail. This enables a clearer identification of the BOEIR Requirements in the Kano model (Maisano, 2025c). Interviews are particularly useful for uncovering Excitement Attributes, since customers may unintentionally tell about features that would distinguish the product.

6.2 House of Quality

Weighting each criterion made it possible to establish a ranking from most to least important; however, it is important to recognise that, because the initial values were defined by the project group, some degree of bias may have been introduced. This limitation could have been mitigated by a peer review conducted by another group. This was not utilized due to the time constraint of the project.

6.3 FMECA

Finally, another limitation is that the identified risks were based primarily on general assumptions rather than asking experienced users. As a result, important failure modes may have been overlooked, and some of the assessed FMECA ratings may have been underestimated or overestimated. The absence of a CAD model and a physical prototype further contributed to this limitation, as practical testing could not be conducted. However, the short duration of the project did not allow for a more comprehensive investigation.

7 Conclusion

This project applied a set of quality engineering tools to support the early design of a robotic vacuum cleaner for families with children and pets. The House of Quality enabled the translation of customer needs into measurable TCs. Benchmarking confirmed strong performance in several areas while revealing improvement opportunities in SLAM update range, airflow rate, and component lifetime.

The Q-Bench optimisation procedure produced a configuration that outperformed existing competitor profiles, demonstrating the value of multi-criteria decision support methods in concept selection. The FMECA complemented this analysis by identifying failure modes.

Several limitations were also identified. Nonetheless, the methods used clearly showed performance priorities and risks, offering a solid foundation for further design iterations and development.

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Appendices

A Path

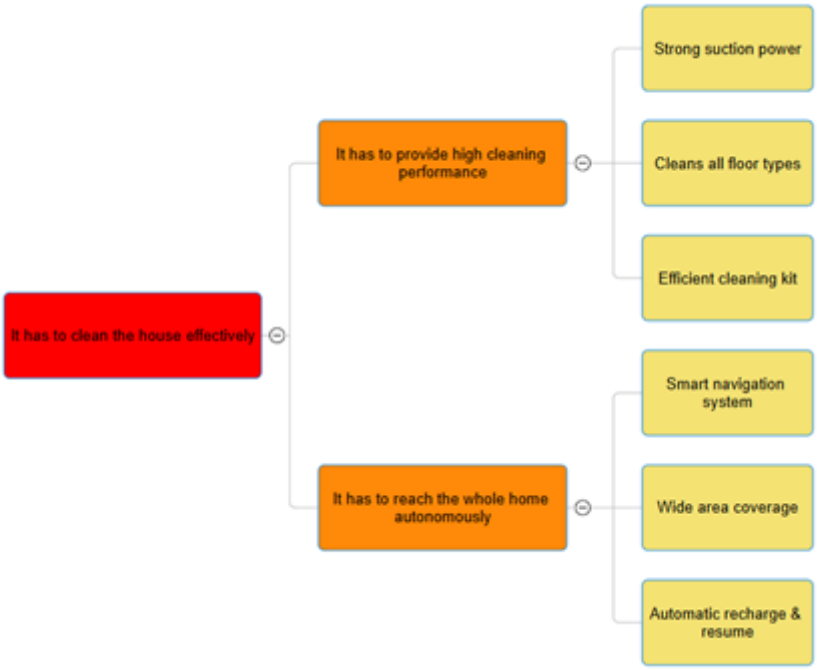

Figure 4: Path 1.

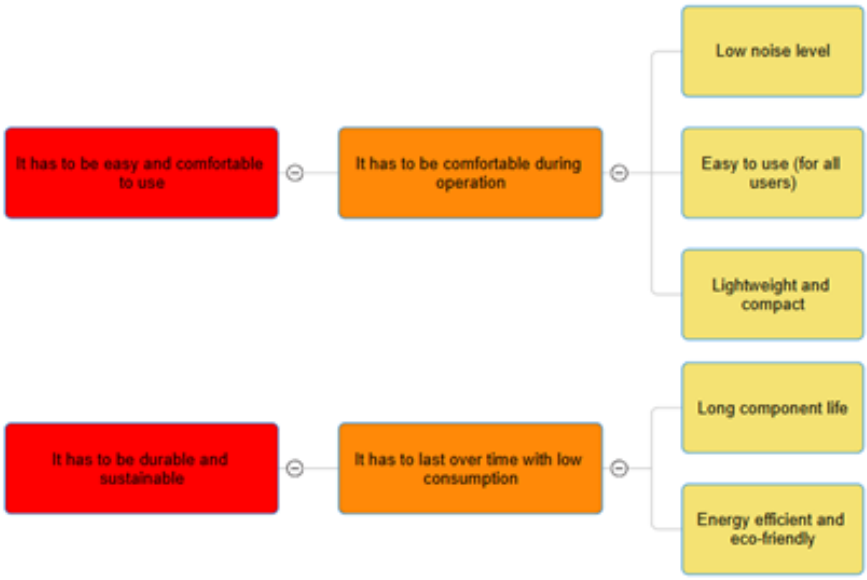

Figure 5: Path 2.

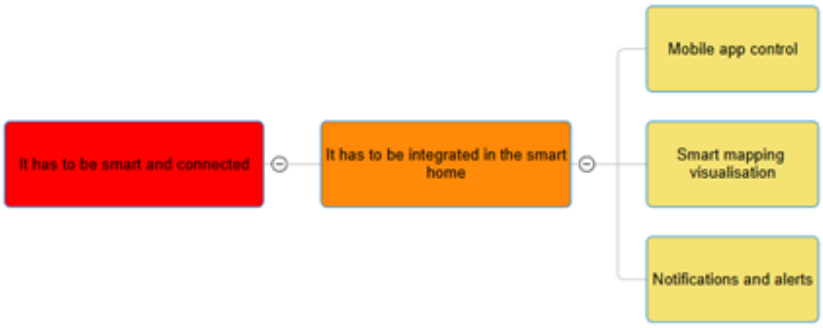

Figure 6: Path 3.

B Histograms

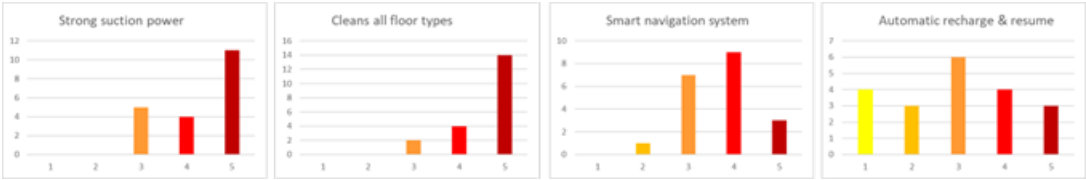

Figure 7: Histogram 1.

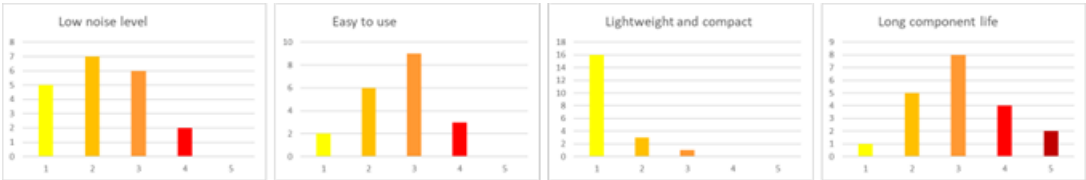

Figure 8: Histogram 2.

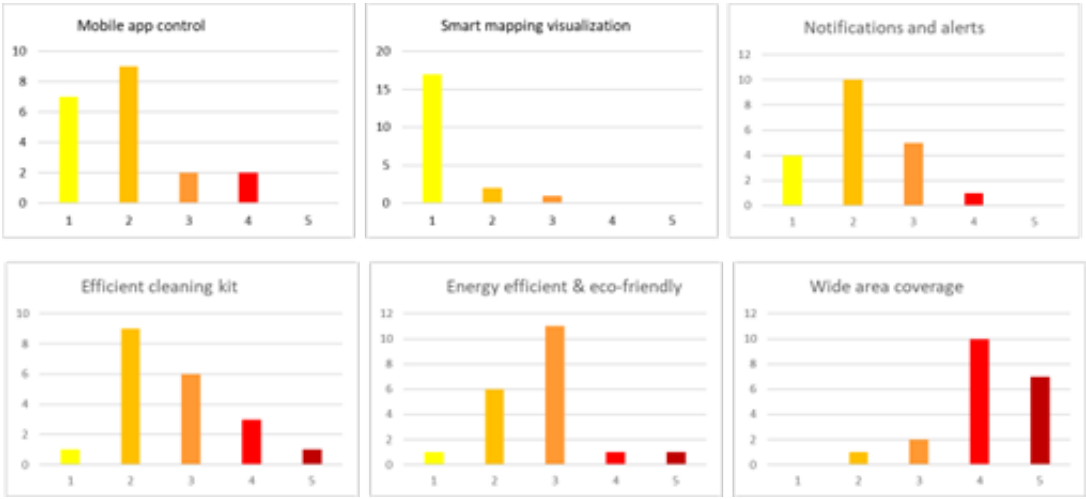

Figure 9: Histogram 3.

C Customer Requirements Weights

Table 4: Customer Requirements.

Customer Requirement	**Degree of Importance**	**Relative Importance**
Strong suction power	5	12%
Cleans all floor types	5	12%
Smart navigation system	4	10%
Automatic recharge & resume	4	10%
Mobile app control	2	5%
Smart mapping visualization	1	2%
Notifications and alerts	2	5%
Low noise level	2	5%
Easy to use (for all users)	3	7%
Lightweight and compact	2	5%
Long component life	3	7%
Efficient cleaning kit	3	7%
Energy efficient and eco-friendly	4	10%
Wide area coverage	3	7%
Total	**41**	**100%**
Mean	**2.93**	

D Customer Requirements Benchmarking

Table 5: Benchmarking of Customer Requirements.

CR	Our Model	Dyson	iRobot	Roborock	Target	Improvement	Strength	Abs. Weight	Rel. Weight
1	4	5	4	4	5	1.25	1.5	9.375	15.32%
2	5	5	5	5	5	1	1.5	7.5	12.26%
3	4	4	5	4	5	1.25	1.2	4.8	7.85%
4	3	5	5	5	4	1.333	1.4	6.54	10.69%
5	3	5	5	5	3	1	1.2	2.4	3.92%
6	3	5	5	5	1	0.333	1.2	1.2	1.96%
7	3	5	5	4	3	1	0.75	2.25	3.61%
8	3	5	4	4	5	1.333	1	3	4.89%
9	3	5	5	4	4	1.333	1	3	4.89%
10	3	4	5	4	3	1	0.75	2.25	3.61%
11	3	4	4	4	3	1	1.5	4.5	7.36%
12	3	5	4	4	4	1.333	1.2	3.6	5.89%
13	3	4	4	4	4	1.333	1.2	3.6	5.89%
14	3	4	4	4	3	1	1.2	3.6	7.85%
Total	47	57	60	62	55			61.175	100.00%

E Formulas

Table 6: Summary of Technical Characteristics, Formulas, and Variable Definitions

Technical Characteristic	**Formula**	**Variable Definitions**
Motor Power (W)	$P = V \times I$	P: power (W); V: voltage (V); I: current (A)
Airflow Rate (m^3/s)	$Q = A \times v$	Q: airflow (m^3/s); A: cross-sectional area (m^2); v: air velocity (m/s)
SLAM Localization Accuracy (cm)	$\text{Accuracy} = \sqrt{\frac{1}{n}\sum(x_i - \hat{x}_i)^2}$	x_i: true position; $\hat{x}_i$: estimated position; n: number of samples
SLAM Update Range (m)	$R_{\max} = \frac{P_t G_t G_r \lambda^2}{(4\pi)^3 S_{\min}}$	P_t: transmitted power; G_t, G_r: antenna gains; λ: wavelength; $S_{\min}$: minimum detectable signal
Path Optimization Metric (%)	—	Metric expresses deviation from an ideal path; higher % = more efficient path
Auto Recharge Time (min)	$t = \frac{C}{I_{\text{charge}}} \times k$	t: charge time (min); C: battery capacity (Ah); I_{charge}: charging current (A); k: efficiency coefficient (1.1–1.3)
Battery Capacity (mAh)	$C = I \times t$ $t_{\text{run}} = \frac{C}{I_{\text{load}}}$	C: capacity (mAh); I: discharge current (A); t: runtime (h); I_{load}: load current (A)
Acoustic Noise Level (dB(A))	$L_p = 20\log_{10}\left(\frac{p}{p_0}\right)$	L_p: sound pressure level (dB); p: measured pressure (Pa); $p_0 = 20\,\mu$Pa: reference pressure
Component Lifetime (Years)	$L = L_0 e^{\frac{E_a}{kT}}$	L: lifetime; L_0: baseline constant; E_a: activation energy; k: Boltzmann constant; T: absolute temperature
Setting Steps (Integer)	$S = \{s_1, s_2, \ldots, s_n\}$	n: number of discrete adjustment steps

F Signs for Relationships

Figure 10: Signs for Relationships.

G Relationship Between CRs and TCs After Lyman's Normalisation.

Table 7: Relationship Between CRs and TCs After Lyman's Normalisation.

CR / TC	Motor power **A**	Airflow rate **B**	SLAM localisation accuracy **C**	SLAM update range **D**	Path optimisation metric **E**	Auto recharge time **F**	Battery capacity **G**	Acoustic noise level **H**	Component lifetime **I**	Setting steps **J**
Strong Suction Power	0.00	2.16	0.00	0.00	0.00	0.72	2.16	2.16	2.16	0.00
Cleans All Floor Types	0.83	2.50	0.00	0.00	0.00	0.00	0.83	0.83	2.50	0.00
Smart Navigation System	0.00	0.00	1.39	1.39	1.39	0.46	0.15	0.00	0.00	0.00
Automatic Recharge & Resume	1.13	0.00	0.38	0.38	0.13	1.13	0.38	0.13	0.38	0.00
Wide Area Coverage	0.00	0.00	0.13	1.16	1.16	0.39	1.16	0.00	0.00	0.00
Energy Efficient and Eco-Friendly	0.73	0.73	0.08	0.24	0.73	0.73	0.73	0.08	0.73	0.00
Low Noise Level	0.50	1.50	0.00	0.00	0.00	0.00	0.00	1.50	0.50	0.00
Easy To Use (for All Users)	0.00	0.00	0.00	0.16	0.48	0.48	1.45	0.00	0.48	1.45
Lightweight and Compact	0.45	0.15	0.00	0.00	0.00	0.00	0.45	0.45	0.00	0.00
Long Component Life	0.40	0.40	0.00	0.00	0.13	1.19	1.19	0.00	1.19	0.00
Mobile App Control	0.00	0.07	0.62	0.07	0.62	0.21	0.00	0.00	0.21	0.62
Smart Mapping Visualisation	0.00	0.00	0.65	0.65	0.22	0.07	0.00	0.00	0.22	0.00
Notifications and Alerts	0.00	0.00	0.18	0.55	0.18	0.55	0.00	0.00	0.55	0.00
Efficient Cleaning Kit	0.00	3.00	0.00	0.00	0.00	0.00	0.00	0.00	3.00	0.00

H Q-Bench

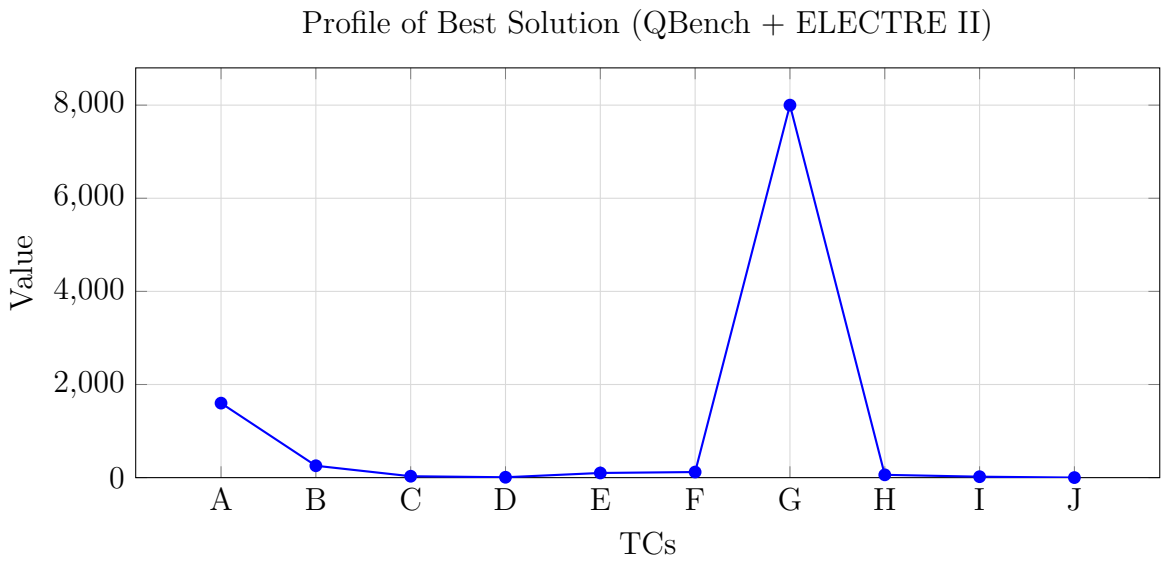

Figure 11: Profile of the Optimised Solution Obtained Using Q-Bench and ELECTRE II.